

Assignment

Name: Arden Diago **Reg No:**

Class: MCA-A

[Repository Link](#)

```
In [23]: import pandas as pd
import numpy as np
from matplotlib import pyplot as plt
```

```
In [24]: DATA='../data/assignment2.csv'
```

```
In [25]: df=pd.read_csv(DATA)
```

```
In [26]: df.head()
```

```
Out[26]:
```

	enrollee_id	city	city_development_index	gender	relevent_experience	enro
0	11541	city_2	0.2	Male	Has relevent experience	
1	11521	city_1	0.2	Male	No relevent experience	
2	11544	city_2	0.2	NaN	No relevent experience	
		city_1	0.2	NaN	No relevent experience	
		city_1	0.2	Male	Has relevent experience	

```
In [27]: df.tail()
```

```
Out[27]:
```

	enrollee_id	city	city_development_index	gender	relevent_experience	enro
11543	2174	city_1	0.2	Male	No relevent experience	
11546	2175	city_1	0.2	Male	Has relevent experience	
11547	2176	city_1	0.2	Male	Has relevent experience	
		city_1	0.2	Male	Has relevent experience	
		city_1	0.2	NaN	No relevent experience	

```
In [28]: df.isnull().mean()*100
```

```
Out[28]: enrollee_id      0.000000
city      0.000000
city_development_index  2.500261
gender    23.530640
relevent_experience    0.000000
enrolled_university    2.014824
education_level        2.401086
major_discipline       14.683161
experience              0.339284
company_size           30.994885
company_type           32.049274
training_hours         3.998330
target                0.000000
dtype: float64
```

```
In [29]: df.shape
```

```
Out[29]: (19158, 13)
```

```
In [30]: cols = [var for var in df.columns if df[var].isnull().mean() < 0.05 and df[var].nunique() > 10]
```

```
Out[30]: ['city_development_index',
'enrolled_university',
'education_level',
'experience',
'training_hours']
```

```
In [31]: df[cols].sample(5)
```

```
Out[31]:
```

	city_development_index	enrolled_university	education_level	experience	training_hours
	.	no_enrollment	Graduate	.	
	.	no_enrollment	Graduate	.	
	.	Full time course	Graduate	.	
	.	no_enrollment	Graduate	.	
	.	no_enrollment	Graduate	.	

```
In [32]: df['education_level'].value_counts()
```

```
Out[32]: education_level
Graduate      11598
Masters        4361
High School    2017
Phd            414
Primary School  308
Name: count, dtype: int64
```

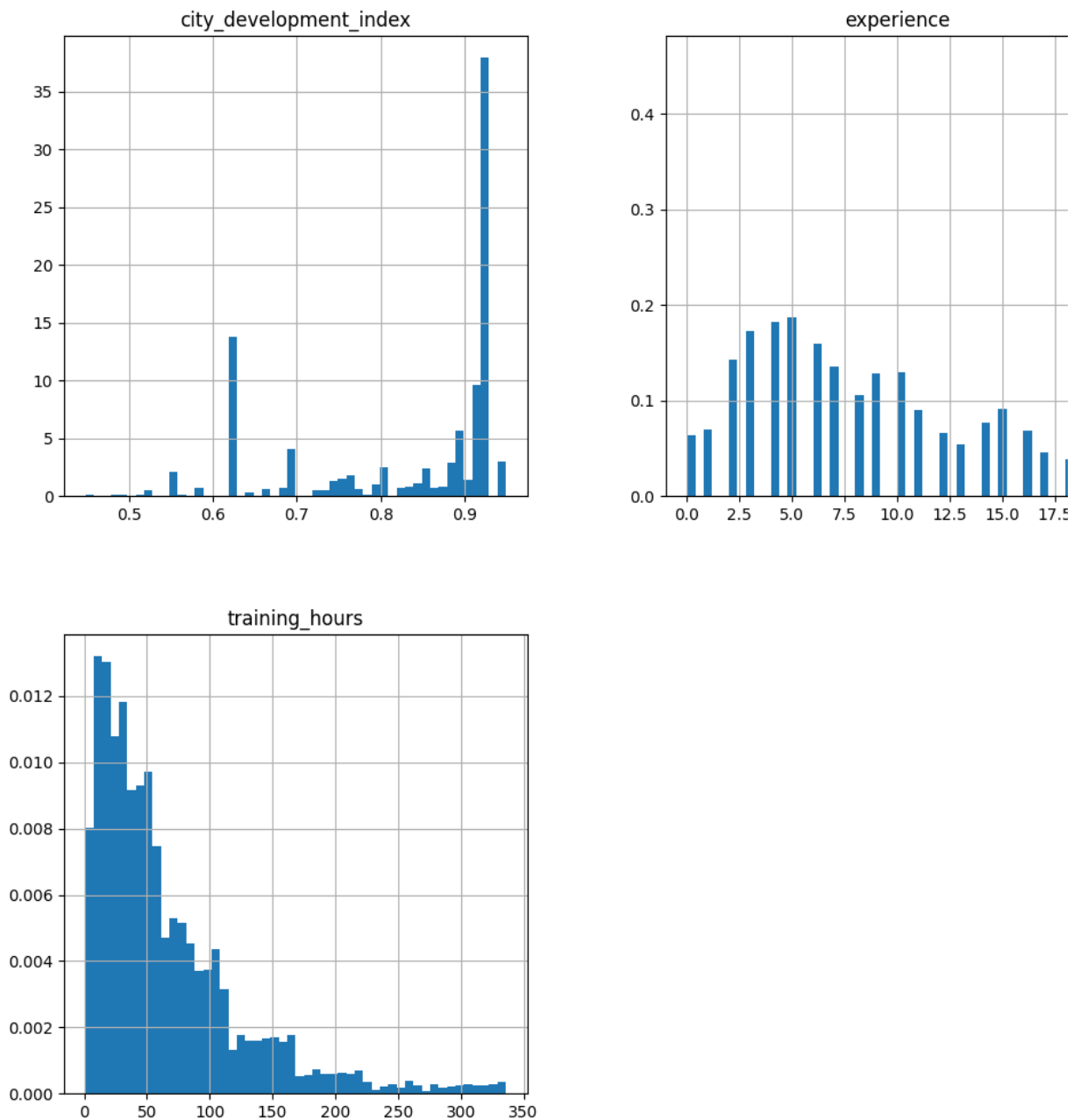
```
In [33]: len(df[cols].dropna()) / len(df)
```

```
Out[33]: 0.8968577095730244
```

```
In [34]: new_df = df[cols].dropna()
df.shape, new_df.shape
```

Out[34]: ((19158, 13), (17182, 5))

```
In [35]: new_df.hist(bins=50, density=True, figsize=(12, 12))
plt.show()
```

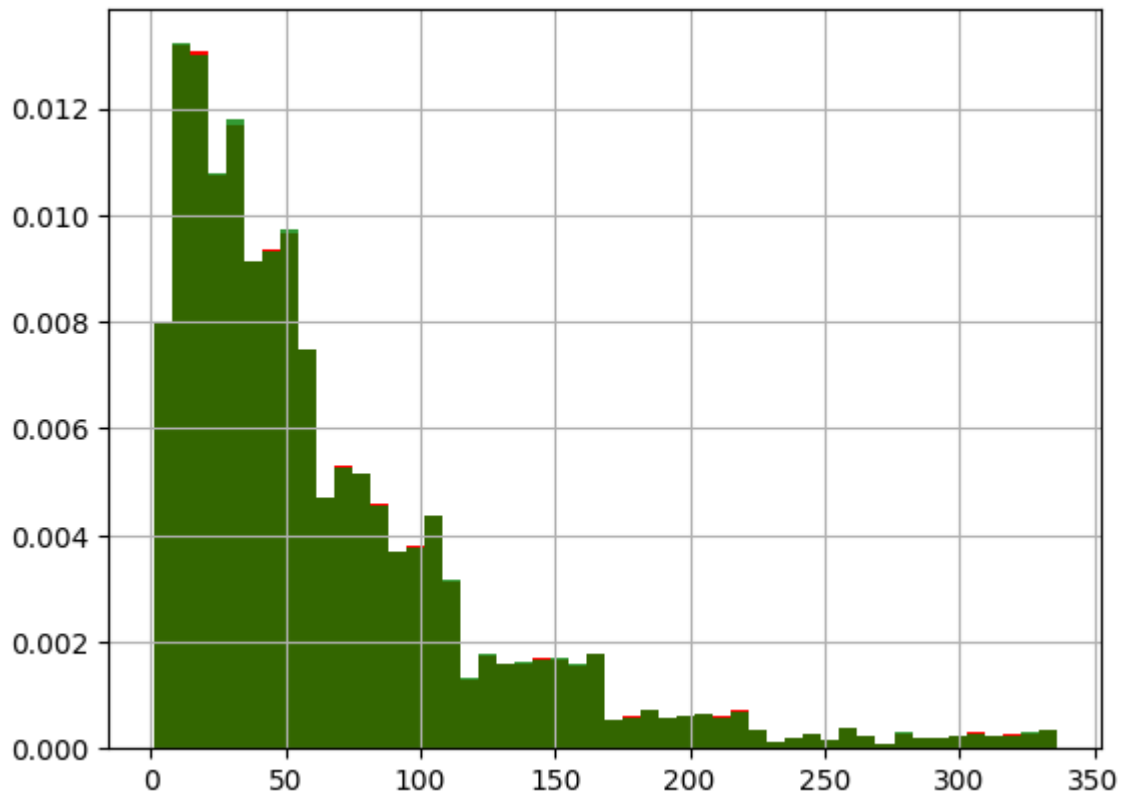


```
In [36]: fig = plt.figure()
ax = fig.add_subplot(111)

# original data
df['training_hours'].hist(bins=50, ax=ax, density=True, color='red')

# data after cca, the argument alpha makes the color transparent, so we c
# see the overlay of the 2 distributions
new_df['training_hours'].hist(bins=50, ax=ax, color='green', density=True
```

Out[36]: <Axes: >

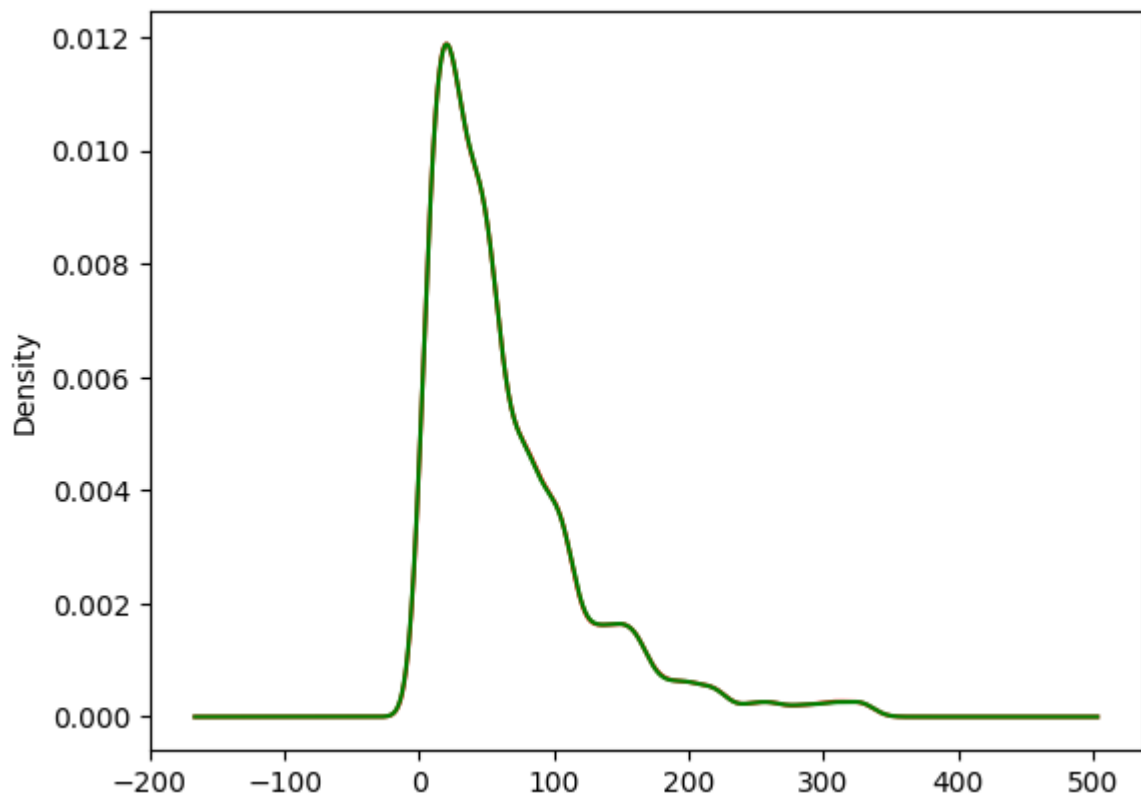


```
In [37]: fig = plt.figure()
ax = fig.add_subplot(111)

# original data
df['training_hours'].plot.density(color='red')

# data after cca
new_df['training_hours'].plot.density(color='green')
```

Out[37]: <Axes: ylabel='Density'>

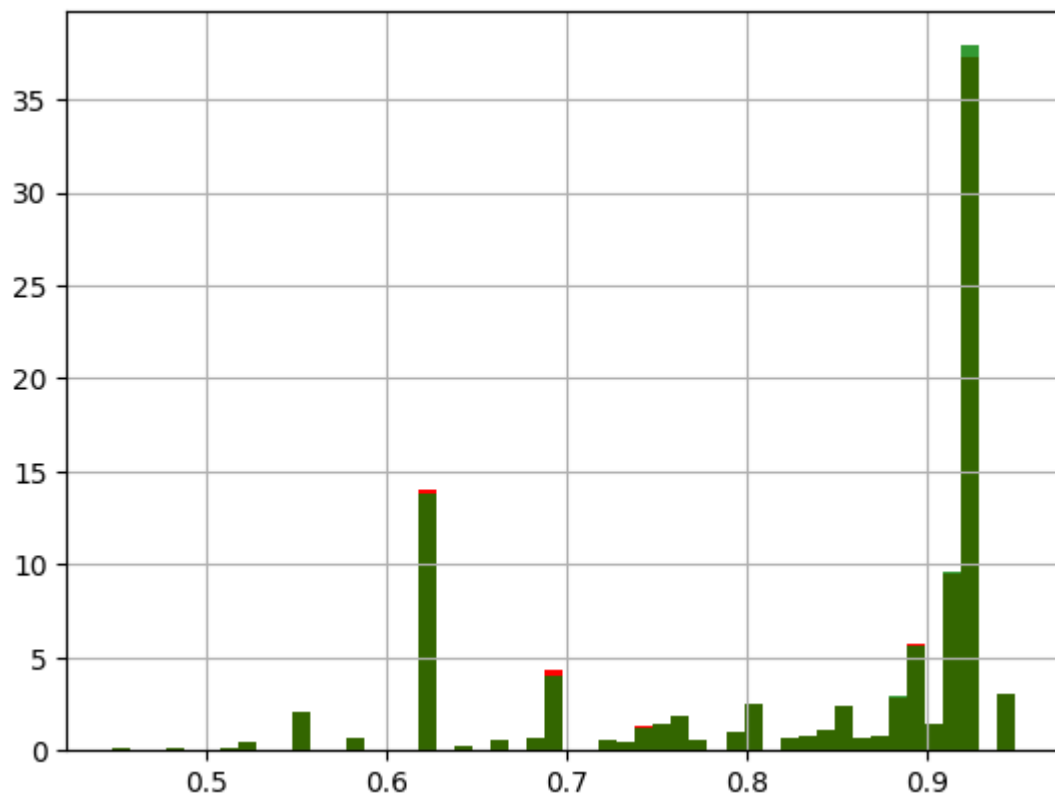


```
In [38]: fig = plt.figure()
ax = fig.add_subplot(111)

# original data
df['city_development_index'].hist(bins=50, ax=ax, density=True, color='red')

# data after cca, the argument alpha makes the color transparent, so we can
# see the overlay of the 2 distributions
new_df['city_development_index'].hist(bins=50, ax=ax, color='green', dens
```

Out[38]: <Axes: >

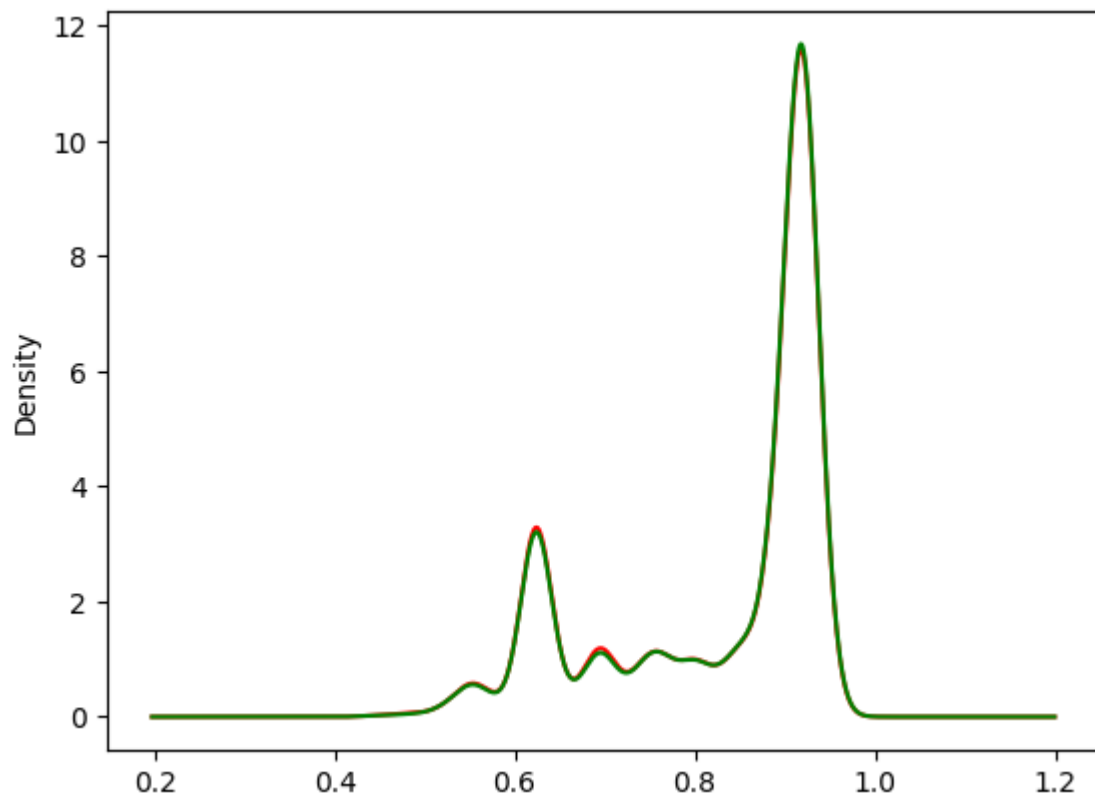


```
In [39]: fig = plt.figure()
ax = fig.add_subplot(111)

# original data
df['city_development_index'].plot.density(color='red')

# data after cca
new_df['city_development_index'].plot.density(color='green')
```

Out[39]: <Axes: ylabel='Density'>

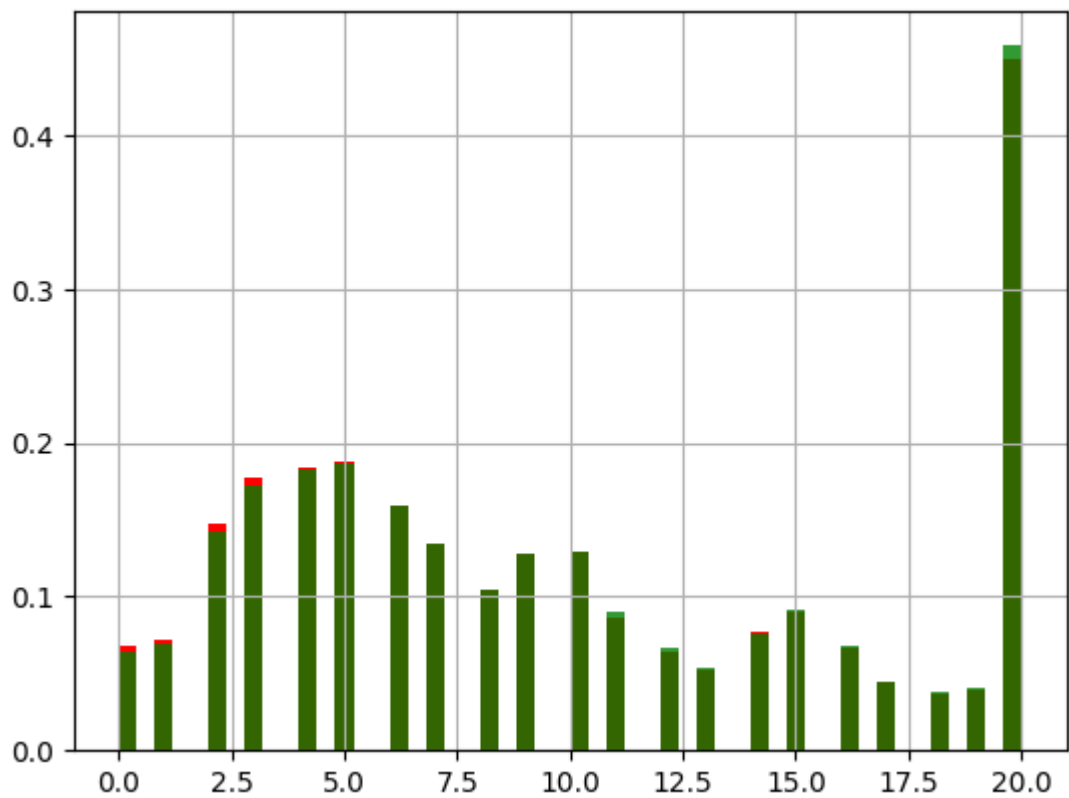


```
In [40]: fig = plt.figure()
ax = fig.add_subplot(111)

# original data
df['experience'].hist(bins=50, ax=ax, density=True, color='red')

# data after cca, the argument alpha makes the color transparent, so we c
# see the overlay of the 2 distributions
new_df['experience'].hist(bins=50, ax=ax, color='green', density=True, al
```

```
Out[40]: <Axes: >
```

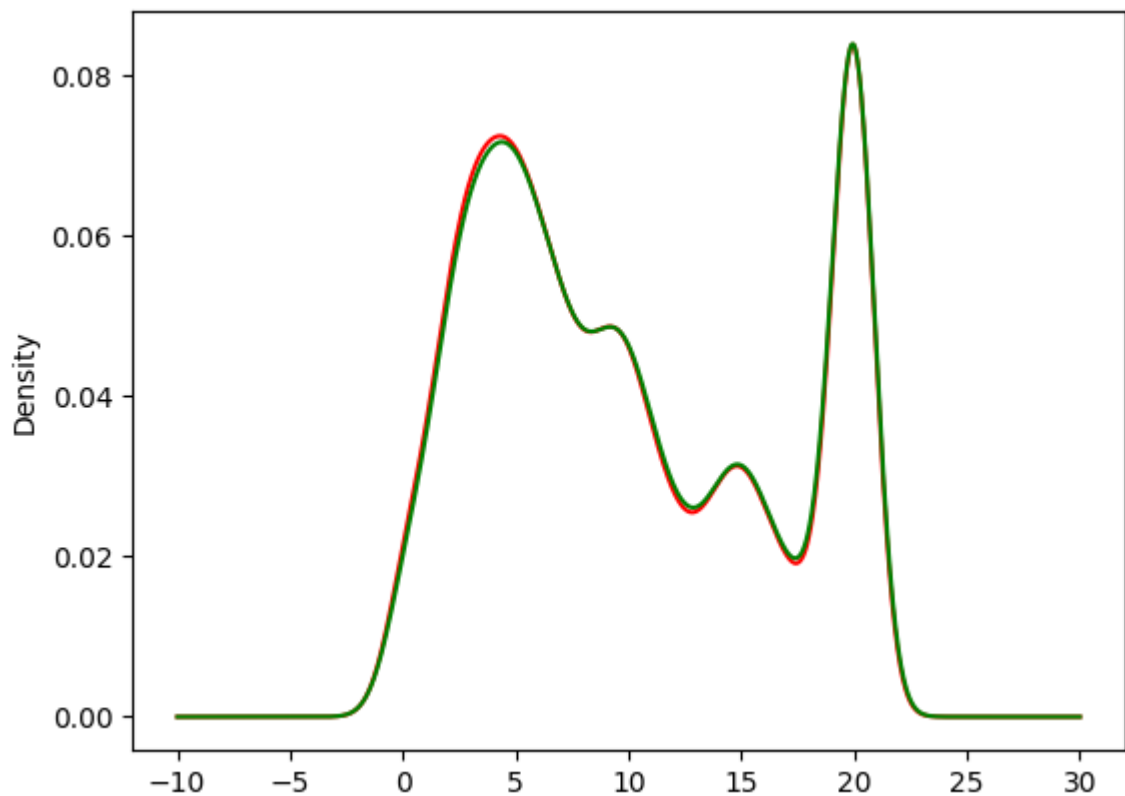


```
In [41]: fig = plt.figure()
ax = fig.add_subplot(111)

# original data
df['experience'].plot.density(color='red')

# data after cca
new_df['experience'].plot.density(color='green')
```

Out[41]: <Axes: ylabel='Density'>



```
In [42]: temp = pd.concat([
    # percentage of observations per category, original data
    df['enrolled_university'].value_counts() / len(df),

    # percentage of observations per category, cca data
    new_df['enrolled_university'].value_counts() / len(new_df)
],
axis=1)

# add column names
temp.columns = ['original', 'cca']

temp
```

```
Out[42]:
```

	original	cca
enrolled_university		
no_enrollment	.	.
Full time course	.	.
Part time course	.	.

```
In [43]: temp = pd.concat([
    # percentage of observations per category, original data
    df['education_level'].value_counts() / len(df),

    # percentage of observations per category, cca data
    new_df['education_level'].value_counts() / len(new_df)
],
axis=1)

# add column names
```

```
temp.columns = ['original', 'cca']  
  
temp
```

Out[43]:

	original	cca
education_level		
Graduate	.	.
Masters	.	.
High School	.	.
Phd	.	.
Primary School	.	.

In []: